

Problem set

Sleep:

```
hasanimam@DESKTOP-GQLSJEU: ~/xv6-labs-2025
$
init: starting sh
$ sixfive sixfive.txt
sixfive sixfive.txt
$
100
10
6
0
$
init: starting sh
$
init: starting sh
$ memdump
Example 1:
01010
0025
Example 2:
$ string
Example 3:
another
Example 4:
$
0
010438967
100
pytz
Example 5:
hello
$
$ echo deadb0de | memdump hcccc
echo deadb0de | memdump hcccc
0200
0507
$
$ echo deadb0de | memdump p
echo deadb0de | memdump p
04010564
$
```

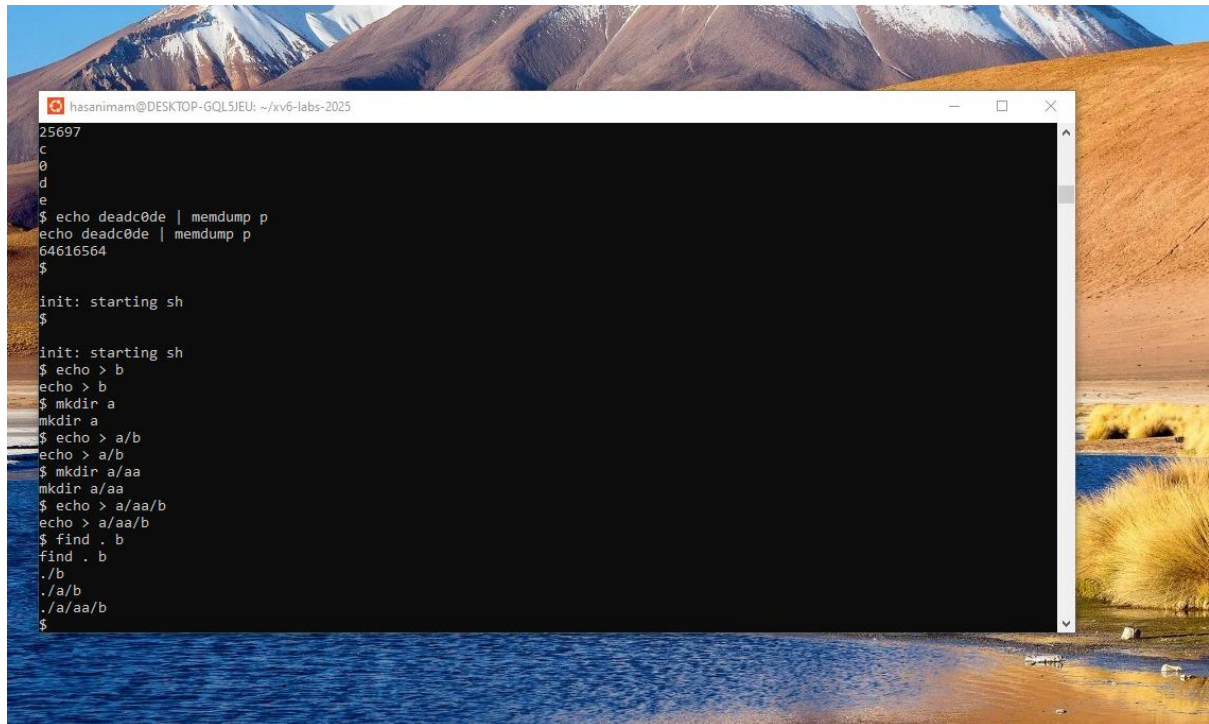
Memdump:

```
hasanimam@DESKTOP-GQLSJEU: ~/xv6-labs-2025
r/printf.o user/umalloc.o
riscv64-unknown-elf-objdump -S user/_forphan > user/forphan.asm
riscv64-unknown-elf-objdump -t user/_forphan | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/forphan.sym
riscv64-unknown-elf-gcc -Wall -Werror -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -DSOL_UTIL -DLAB_UTIL -MD -mcmmodel=media
ny -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memse
t -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fn
o-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fpri
ntf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o user/dorphan.o user/dorphan.c
riscv64-unknown-elf-ld -z max-page-size=4096 -T user/user.ld -o user/_dorphan user/dorphan.o user/ulib.o user/usys.o use
r/printf.o user/umalloc.o
riscv64-unknown-elf-objdump -S user/_dorphan > user/dorphan.asm
riscv64-unknown-elf-objdump -t user/_dorphan | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/dorphan.sym
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo user/_find user/_memdump user/_time user/
_sixfive user/_sleep user/_uptime user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm
user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan
user/_memdump
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
ballocc: first 1222 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-1
egacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

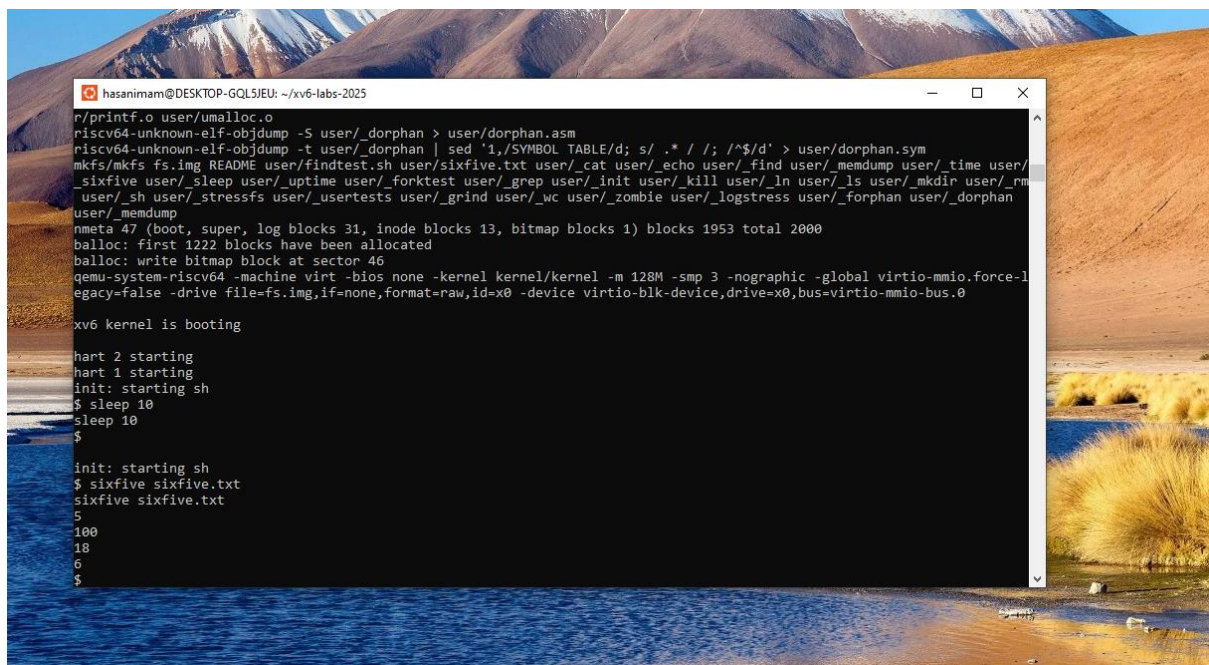
hart 2 starting
hart 1 starting
init: starting sh
$ sleep 10
sleep 10
$
```

Find:



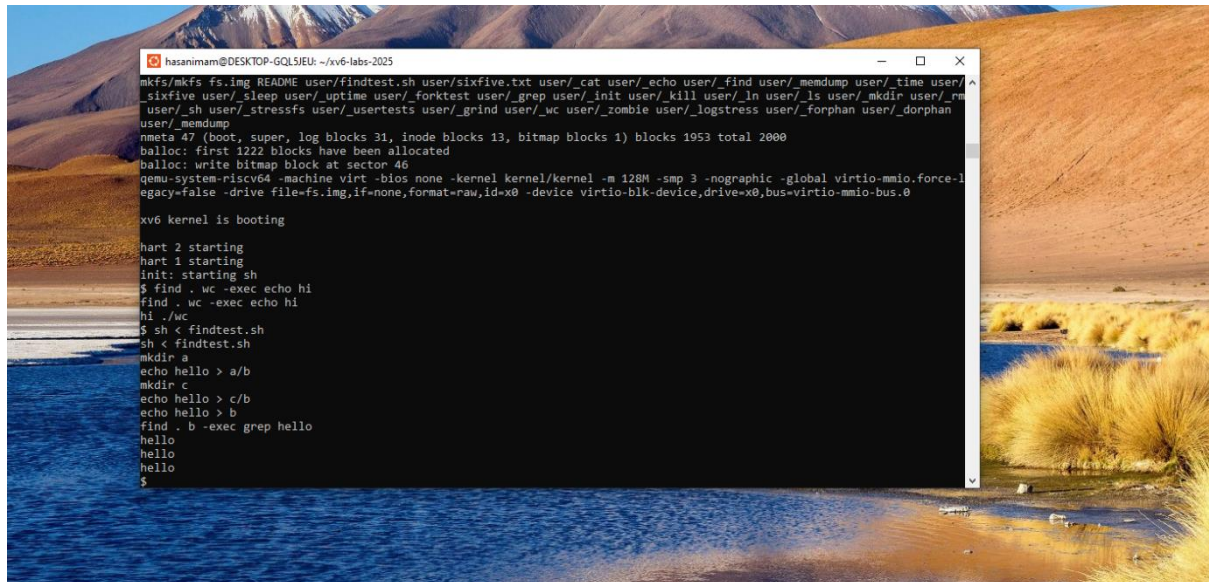
```
hasanimam@DESKTOP-GQL5JEU: ~/xv6-labs-2025
25697
c
0
d
e
$ echo deadc0de | memdump p
echo deadc0de | memdump p
64616564
$
init: starting sh
$
init: starting sh
$ echo > b
echo > b
$ mkdir a
mkdir a
$ echo > a/b
echo > a/b
$ mkdir a/aa
mkdir a/aa
$ echo > a/aa/b
echo > a/aa/b
$ find . b
find . b
./b
./a/b
./a/aa/b
$
```

Sixfive:



```
hasanimam@DESKTOP-GQL5JEU: ~/xv6-labs-2025
r/printf.o user/umalloc.o
riscv64-unknown-elf-objdump -S user/_dorphan > user/dorphan.asm
riscv64-unknown-elf-objdump -t user/_dorphan | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/dorphan.sym
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo user/_find user/_memdump user/_time user/_sleep user/_uptime user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertestfs user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_memdump
nmtest 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
ballocc: first 1222 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0
xv6 kernel is booting
hart 2 starting
hart 1 starting
init: starting sh
$ sleep 10
sleep 10
$
init: starting sh
$ sixfive sixfive.txt
sixfive sixfive.txt
5
100
18
6
$
```

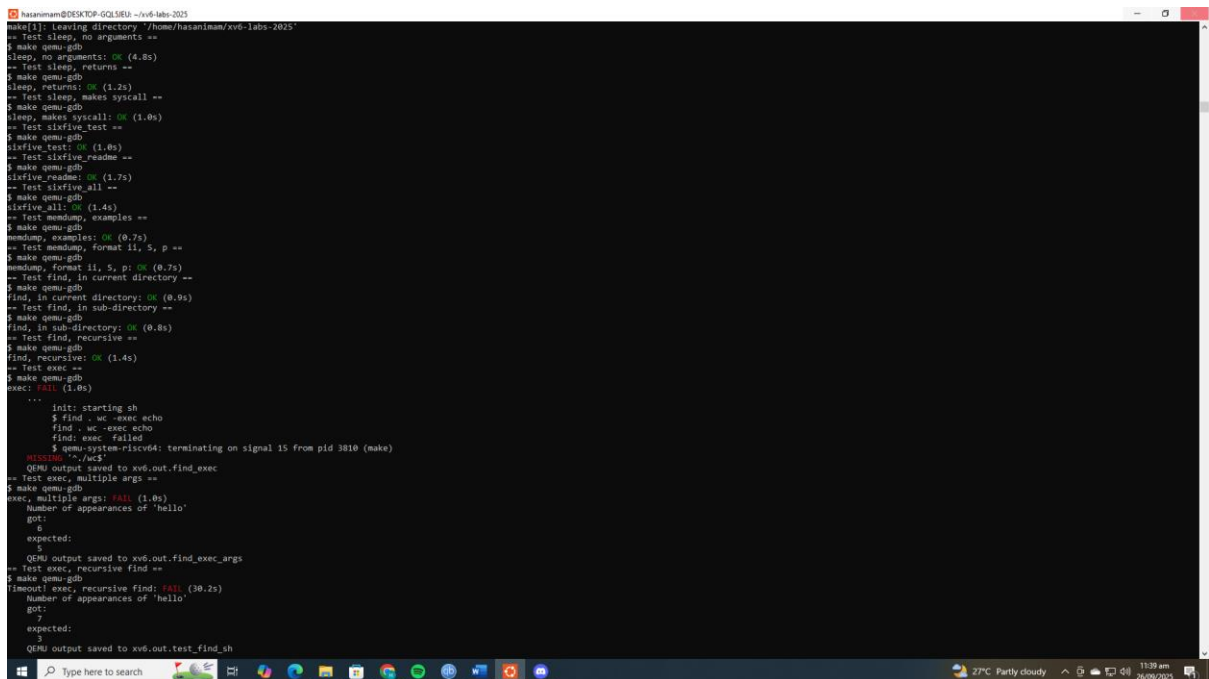
Exec:



```
hasanimam@DESKTOP-GQLSJEU: ~/xv6-labs-2025
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo user/_find user/_mempdump user/_time user/_
sixfive user/_sleep user/_uptime user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm
user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan
user/_mempdump
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
ballocc: first 1222 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting
hart 2 starting
hart 1 starting
init: starting sh
$ find . -wc -exec echo hi
find . -wc -exec echo hi
hi ./wc
$ sh < findtest.sh
sh < findtest.sh
mkdir a
echo hello > a/b
mkdir c
echo hello > c/b
echo hello > b
find . -b -exec grep hello
hello
hello
hello
$
```

Make grade



```
hasanimam@DESKTOP-GQLSJEU: ~/xv6-labs-2025
make[1]: Leaving directory '/home/hasanimam/xv6-labs-2025'
-- Test sleep, no arguments ==
$ make qemu-gdb
sleep, no arguments: OK (4.8s)
-- Test sleep, returns ==
$ make qemu-gdb
sleep, returns: OK (1.2s)
-- Test sleep, makes syscall ==
$ make qemu-gdb
sleep, makes syscall: OK (1.0s)
-- Test sixfive test ==
$ make qemu-gdb
sixfive test: OK (1.0s)
-- Test sixfive_readme ==
$ make qemu-gdb
sixfive_readme: OK (1.7s)
-- Test sixfive_all ==
$ make qemu-gdb
sixfive_all: OK (1.4s)
-- Test memdump, examples ==
$ make qemu-gdb
memdump, examples: OK (0.7s)
-- Test memdump, format ii, s, p ==
$ make qemu-gdb
memdump, format ii, s, p: OK (0.7s)
-- Test find, in current directory ==
$ make qemu-gdb
find, in current directory: OK (0.0s)
-- Test find, in sub-directory ==
$ make qemu-gdb
find, in sub-directory: OK (0.6s)
-- Test find, recursive ==
$ make qemu-gdb
find, recursive: OK (1.4s)
-- Test exec ==
$ make qemu-gdb
exec: FAIL (1.0s)
...
init: starting sh
$ find . -wc -exec echo
find . -wc -exec echo
find: exec: failed
$ qemu-system-riscv64: terminating on signal 15 from pid 3810 (make)
RESIGNAL: "/.wc3"
QEMU output saved to xv6.out.find_exec
-- Test exec, multiple args ==
$ make qemu-gdb
exec, multiple args: FAIL (1.0s)
Number of appearances of 'hello'
got:
0
expected:
3
QEMU output saved to xv6.out.find_exec_args
-- Test exec, recursive find ==
$ make qemu-gdb
timeout! exec, recursive find: FAIL (30.2s)
Number of appearances of 'hello'
got:
7
expected:
3
QEMU output saved to xv6.out.test_find_sh
```

```
hasaninam@DESKTOP-GQL53EU: ~/xv6-labs-2025
sleep, returns: OK (1.2s)
== Test sleep, makes syscall ==
$ make qemu-gdb
sleep, makes syscall: OK (1.0s)
== Test sixfive_test ==
$ make qemu-gdb
sixfive_test: OK (1.0s)
== Test sixfive_readme ==
$ make qemu-gdb
sixfive_readme: OK (1.7s)
== Test sixfive_all ==
$ make qemu-gdb
sixfive_all: OK (1.4s)
== Test memdump, examples ==
$ make qemu-gdb
memdump, examples: OK (0.7s)
== Test memdump, format ii, S, p ==
$ make qemu-gdb
memdump, format ii, S, p: OK (0.7s)
== Test find, in current directory ==
$ make qemu-gdb
find, in current directory: OK (0.9s)
== Test find, in sub-directory ==
$ make qemu-gdb
find, in sub-directory: OK (0.8s)
== Test find, recursive ==
$ make qemu-gdb
find, recursive: OK (1.4s)
== Test exec ==
$ make qemu-gdb
exec: FAIL (1.0s)
...
init: starting sh
$ find . wc -exec echo
find . wc -exec echo
find: exec failed
$ qemu-system-riscv64: terminating on signal 15 from pid 3810 (make)
MISSING '^./wc$'
QEMU output saved to xv6.out.find_exec
== Test exec, multiple args ==
$ make qemu-gdb
exec, multiple args: FAIL (1.0s)
Number of appearances of 'hello'
got:
6
expected:
5
QEMU output saved to xv6.out.find_exec_args
== Test exec, recursive find ==
$ make qemu-gdb
[timeout] exec, recursive find: FAIL (30.2s)
Number of appearances of 'hello'
got:
7
expected:
3
QEMU output saved to xv6.out.test_find_sh
== Test time ==
time: FAIL
Cannot read time.txt
Score: 100/131
make: *** [Makefile:364: grade] Error 1
hasaninam@DESKTOP-GQL53EU: ~/xv6-labs-2025$
```

```
hasaninam@DESKTOP-GQL53EU: ~/xv6-labs-2025
== Test memdump, examples ==
$ make qemu-gdb
memdump, examples: OK (0.7s)
== Test memdump, format ii, S, p ==
$ make qemu-gdb
memdump, format ii, S, p: FAIL (1.3s)
...
$ echo abcdefgh12345678 | memdump ii
echo abcdefgh12345678 | memdump ii
0000 16842440
0000 1751008885
$ echo abcdefgh12345678 | memdump S
echo abcdefgh12345678 | memdump S
0000 abcdefgh12345678
$ echo abcdefgh12345678 | memdump p
echo abcdefgh12345678 | memdump p
00007660504030201
$ qemu-system-riscv64: terminating on signal 15 from pid 11684 (make)
MISSING '"64636261"'
QEMU output saved to xv6.out.memdump_examples
== Test find, in current directory ==
$ make qemu-gdb
find, in current directory: OK (1.0s)
== Test find, in sub-directory ==
$ make qemu-gdb
find, in sub-directory: OK (1.0s)
== Test find, recursive ==
$ make qemu-gdb
find, recursive: OK (1.3s)
== Test exec ==
$ make qemu-gdb
exec: OK (0.8s)
== Test exec, multiple args ==
$ make qemu-gdb
exec, multiple args: FAIL (1.0s)
Number of appearances of 'hello'
got:
6
expected:
5
QEMU output saved to xv6.out.find_exec_args
== Test exec, recursive find ==
$ make qemu-gdb
[timeout] exec, recursive find: FAIL (30.3s)
Number of appearances of 'hello'
got:
7
expected:
3
QEMU output saved to xv6.out.test_find_sh
== Test time ==
time: FAIL
Cannot read time.txt
Score: 100/131
make: *** [Makefile:362: grade] Error 1
hasaninam@DESKTOP-GQL53EU: ~/xv6-labs-2025$
```